# Lecture 8 Exercise Solutions

1. Is $\bar{3} \in Z_{1009}^{\times}$ a square?

   1009 is prime.

   $$\left(\frac{3}{1009}\right) = x \cdot \left(\frac{1009}{3}\right)$$

   $3 \equiv 3 \bmod 4 \wedge 1009 \equiv 1 \bmod 4 \Rightarrow x = 1$

   $1009 \equiv 1 \bmod 3$

   $$\therefore \left(\frac{3}{1009}\right) = \left(\frac{1}{3}\right) = 1$$

   Since 1 is always a square of itself, $\bar{3} \in Z_{1009}^{\times}$ is a square.

2. Is $\overline{-5} \in Z_{1009}^{\times}$ a square?

   $$\left(\frac{-5}{1009}\right) = \left(\frac{-1}{1009}\right)\left(\frac{5}{1009}\right)$$

   $$1009 \equiv 1 \bmod 4 \Rightarrow \left(\frac{-1}{1009}\right) = 1$$

   $$\left(\frac{5}{1009}\right) = x \cdot \left(\frac{1009}{5}\right)$$

   $1009 \equiv 1 \bmod 4 \wedge 5 \equiv 1 \bmod 4 \Rightarrow x = 1$

   $1009 \equiv 4 \bmod 5$

   $$\therefore \left(\frac{-5}{1009}\right) = \left(\frac{-1}{1009}\right)\left(\frac{5}{1009}\right) = \left(\frac{1009}{5}\right) = \left(\frac{4}{5}\right) = \left(\frac{2}{5}\right)^2$$

   $$5 \equiv 5 \bmod 8 \Rightarrow \left(\frac{2}{5}\right) = -1$$

   $$\left(\frac{-5}{1009}\right) = \left(\frac{2}{5}\right)^2 = (-1)^2 = 1$$

   So $\overline{-5} \in Z_{1009}^{\times}$ is a square.

3. Evaluate $\left(\frac{-1}{2033}\right)$

$2033 = 19 \cdot 107$

$\therefore \left(\frac{-1}{2033}\right) = \left(\frac{-1}{19}\right)\left(\frac{-1}{107}\right)$

$19 \equiv 3 \bmod 4 \Rightarrow \left(\frac{-1}{19}\right) = -1$

$107 \equiv 3 \bmod 4 \Rightarrow \left(\frac{-1}{107}\right) = -1$

$\therefore \left(\frac{-1}{2033}\right) = \left(\frac{-1}{19}\right)\left(\frac{-1}{107}\right) = (-1)(-1) = 1$

Note that -1 is a square in neither $Z_{19}^{\times}$ nor $Z_{107}^{\times}$. But, nevertheless $\left(\frac{-1}{2033}\right) = 1$.

It is one of those situations where the Jacobi symbol equal to 1 does not tell us whether -1 is a square or not. It is not in this case because its not a square in one of the component groups.

4. Evaluate $\left(\frac{-1}{2041}\right)$

$2041 = 13 \cdot 157$

$\therefore \left(\frac{-1}{2041}\right) = \left(\frac{-1}{13}\right)\left(\frac{-1}{157}\right)$

$13 \equiv 1 \bmod 4 \Rightarrow \left(\frac{-1}{13}\right) = 1$

$157 \equiv 1 \bmod 4 \Rightarrow \left(\frac{-1}{157}\right) = 1$

$\therefore \left(\frac{-1}{2041}\right) = \left(\frac{-1}{13}\right)\left(\frac{-1}{157}\right) = (1)(1) = 1$

5. Evaluate $\left(\frac{119}{2009}\right)$

We can't compute this because $\gcd(119, 2009) = 7 \Rightarrow 199 \notin Z_{2009}^{\times}$.

6. Evaluate $\left(\frac{46}{111}\right)$

$111 = 3 \cdot 37$

$\therefore \left(\frac{46}{111}\right) = \left(\frac{46}{3}\right)\left(\frac{46}{37}\right) = \left(\frac{2}{3}\right)\left(\frac{23}{3}\right)\left(\frac{2}{37}\right)\left(\frac{23}{37}\right)$

$3 \equiv 3 \bmod 8 \Rightarrow \left(\frac{2}{3}\right) = -1$

$37 \equiv 5 \bmod 8 \Rightarrow \left(\frac{2}{37}\right) = -1$

$\therefore \left(\frac{46}{111}\right) = \left(\frac{46}{3}\right)\left(\frac{46}{37}\right) = (-1)\left(\frac{23}{3}\right)(-1)\left(\frac{23}{37}\right) = \left(\frac{23}{3}\right)\left(\frac{23}{37}\right)$

$23 \equiv 2 \bmod 3$

$\therefore \left(\frac{46}{111}\right) = \left(\frac{2}{3}\right)\left(\frac{23}{37}\right)$

$3 \equiv 3 \bmod 8 \Rightarrow \left(\frac{2}{3}\right) = -1$

$\therefore \left(\frac{46}{111}\right) = -\left(\frac{23}{37}\right) = -x \cdot \left(\frac{37}{23}\right)$

$23 \equiv 3 \bmod 4 \wedge 37 \equiv 1 \bmod 4 \Rightarrow x = 1$

$37 \equiv 14 \bmod 23$

$\therefore \left(\frac{46}{111}\right) = -\left(\frac{23}{37}\right) = -\left(\frac{14}{23}\right) = -\left(\frac{2}{23}\right)\left(\frac{7}{23}\right)$

$23 \equiv 7 \bmod 8 \Rightarrow \left(\frac{2}{23}\right) = 1$

$\therefore \left(\frac{46}{111}\right) = -\left(\frac{23}{37}\right) = -\left(\frac{14}{23}\right) = -\left(\frac{7}{23}\right) = -x \cdot \left(\frac{23}{7}\right)$

$7 \equiv 3 \bmod 4 \wedge 23 \equiv 3 \bmod 4 \Rightarrow x = -1$

$23 \equiv 2 \bmod 7$

$\therefore \left(\frac{46}{111}\right) = -\left(\frac{23}{37}\right) = -\left(\frac{14}{23}\right) = -\left(\frac{7}{23}\right) = -(-1) \cdot \left(\frac{2}{7}\right) = \left(\frac{2}{7}\right)$

$7 \equiv 7 \bmod 8 \Rightarrow \left(\frac{2}{7}\right) = 1$

$\therefore \left(\frac{46}{111}\right) = -\left(\frac{23}{37}\right) = -\left(\frac{14}{23}\right) = -\left(\frac{7}{23}\right) = \left(\frac{2}{7}\right) = 1$

7. Evaluate $\left(\frac{37}{112}\right)$

$112 = 7 \cdot 2^4$

$\therefore \left(\frac{37}{112}\right) = \left(\frac{37}{7}\right)\left(\frac{37}{2}\right)^4$

Whether $\left(\frac{37}{2}\right) = 1$ or $\left(\frac{37}{2}\right) = -1$, $\left(\frac{37}{2}\right)^4 = 1$

$\therefore \left(\frac{37}{112}\right) = \left(\frac{37}{7}\right)$

$37 \equiv 2 \bmod 7$

$\therefore \left(\frac{37}{112}\right) = \left(\frac{2}{7}\right)$

$7 \equiv 7 \bmod 8$

$\therefore \left(\frac{37}{112}\right) = \left(\frac{2}{7}\right) = 1$

8. Evaluate $\left(\frac{113}{2462}\right)$

$2462 = 2 \cdot 1231$

$\therefore \left(\frac{113}{2462}\right) = \left(\frac{113}{2}\right)\left(\frac{113}{1231}\right)$

$113 \bmod 2 = 1$

$\therefore \left(\frac{113}{2462}\right) = \left(\frac{1}{2}\right)\left(\frac{113}{1231}\right)$

But 1 is always a square of itself so $\left(\frac{1}{2}\right) = 1$

$\therefore \left(\frac{113}{2462}\right) = \left(\frac{113}{1231}\right) = x \cdot \left(\frac{1231}{113}\right)$

$113 \equiv 1 \bmod 4 \wedge 1231 \equiv 3 \bmod 4 \Rightarrow x = 1$

$1231 \equiv 101 \bmod 113$

$\therefore \left(\frac{113}{2462}\right) = \left(\frac{113}{1231}\right) = \left(\frac{101}{113}\right) = x \cdot \left(\frac{112}{101}\right)$

$101 \equiv 3 \bmod 4 \wedge 113 \equiv 1 \bmod 4 \Rightarrow x = 1$

$112 \equiv 11 \bmod 101$

$\therefore \left(\frac{113}{2462}\right) = \left(\frac{113}{1231}\right) = \left(\frac{101}{113}\right) = \left(\frac{11}{101}\right) = x \cdot \left(\frac{101}{11}\right)$

$11 \equiv 1 \bmod 4 \wedge 101 \equiv 1 \bmod 4 \Rightarrow x = 1$

$101 \equiv 2 \bmod 11$

$\therefore \left(\frac{113}{2462}\right) = \left(\frac{113}{1231}\right) = \left(\frac{101}{113}\right) = \left(\frac{11}{101}\right) = \left(\frac{2}{11}\right)$

$11 \equiv 3 \bmod 8 \Rightarrow \left(\frac{2}{11}\right) = -1$

$\therefore \left(\frac{113}{2462}\right) = \left(\frac{113}{1231}\right) = \left(\frac{101}{113}\right) = \left(\frac{11}{101}\right) = \left(\frac{2}{11}\right) = -1$